

# Temperature Robustness Is a Model Property, Not a Sampler Choice

Anonymous ACL submission

## Abstract

Truncation samplers such as top- $p$ , min- $p$ , and top- $n\sigma$  exist to prevent the accuracy collapse of high-temperature sampling. Sampler evaluations compare one rule against another and take the collapse for granted. We ask whether it occurs in every model at temperatures used in deployment ( $T \leq 1.3$ ). In a single controlled pipeline, we run seven instruction-tuned models from five families across 8 sampler arms, 4 quantization levels, and 2 tasks. In that range, sampler choice barely matters in every family except Llama. Under plain temperature sampling at  $T=1.3$ , Llama-3.1-8B and Llama-3.2-3B lose 34 to 41 accuracy points on GSM8K and MMLU-Pro, and Llama-3-8B loses 17 on MMLU-Pro; no model outside the family loses more than 8. The collapse is a termination failure. Generations derail midway and never reach an answer, but when a model does produce a well-formed answer, it is as accurate as under greedy decoding. A temperature ladder up to  $T=2.0$  shows that every model collapses at some temperature, and truncation samplers begin to matter only past it. Forcing one off-distribution token into the context multiplies a fragile model’s chance of drawing another, so errors compound. Practitioners tuning a current model can leave the sampler alone.

## 1 Introduction

Sampling is the default decoding choice for deployed language models (Shi et al., 2024), and truncation samplers exist to make it safe at high temperature. They were designed for open-ended generation, where output variety is the goal (Garces Arias et al., 2024), but they are now also evaluated on tasks with one checkable answer. Min- $p$  reports accuracy gains over top- $p$  on GSM8K at high temperature (Nguyen et al., 2025), and top- $n\sigma$  reports accuracy maintained up to  $T=3.0$  (Tang et al., 2024). Both results rest on the same premise, that raising the temperature eventually ruins every model’s accuracy and a good sampler limits the damage. We

test that premise at the temperatures deployments actually use and ask **whether the collapse is a property of decoding, or of the model.**

There is a good reason to expect the collapse. At each step the decoder draws the next token at random from the model’s predicted distribution, and a temperature parameter scales the randomness of the draw. Raising it moves probability from the model’s preferred tokens toward the rare ones in the tail of the distribution, where the model’s probability estimates are least reliable, so a high-temperature draw can select a token that fits the context poorly (Holtzman et al., 2020; Hewitt et al., 2022). The selected token then becomes context for every later step, and the continuation can degenerate into incoherent text. Truncation samplers prevent this degeneration by deleting the tail of the distribution before the draw, and the rules differ in where they cut. Top- $k$ , nucleus (top- $p$ ), min- $p$ , and top- $n\sigma$  are the four we test, and Section 2 describes each.

The accuracy evidence has two problems. First, it generalizes from very few models. Top- $n\sigma$ ’s results come from a single model, Llama-3-8B (Tang et al., 2024), and min- $p$ ’s main results from Mistral-7B (Nguyen et al., 2025). Second, the gains are demonstrated at temperatures of 1.5 to 3, while deployed systems ship defaults between 0.6 and 1.0 (Section 3). Schaeffer et al. (2025) reanalyzed min- $p$  and found the samplers hard to separate once each one receives an equal tuning budget, but the reanalysis still asks which sampler wins, and leaves the premise itself untouched.

Across nine instruction-tuned models from five families, run through one controlled pipeline, the answer is the model. The six models outside the Llama family hold their accuracy at deployment temperatures ( $T \leq 1.3$ ) under every sampler we test, staying within 8 points of their low-temperature scores. Llama-3.1-8B and Llama-3.2-3B lose 34 to 41 accuracy points under plain temperature sam-

pling, and Llama-3-8B, top- $n\sigma$ 's test model, loses 17 on the longer task.

Our contributions:

- **A single-pipeline factorial study.** Seven instruction-tuned models, four GGUF quantization levels plus a BF16 anchor, six decoding methods plus two ablation arms that swap the order in which temperature and truncation are applied, and  $T \in \{0.7, 1.0, 1.3\}$ , on GSM8K and MMLU-Pro ( $N=50$  items,  $K=3$  repetitions): about 204,000 generations, with item-clustered bootstrap confidence intervals throughout.
- **Robustness is model-bound, and the fragility runs in one family.** Every collapse we observe is a Llama, across three of its generations (Section 4.1); on the robust models the sampler is worth at most a few points, and even those overstate the true differences (Section 4.2).
- **The collapse is a termination failure.** Conditional on emitting a well-formed final answer, accuracy is flat in temperature everywhere. What varies is the probability of derailing into unparseable text mid-generation (Section 4.3).
- **A cliff map from  $T=1.3$  to 2.0.** Every model collapses somewhere; what distinguishes models is where. On GSM8K, Llama collapses by  $T=1.5$ , Qwen2.5 by 1.7, and Gemma-3 by 2.0. Truncation samplers shift each cliff right in order of tail-cutting strength, so the gains the sampler papers report are real and sit past the cliff, above deployment temperatures (Section 4.4).
- **A mechanism signature.** A logit probe and a forced-token experiment locate the fragility after the first off-distribution token enters the context. One injected tail token multiplies Llama's chance of drawing another 2.3-fold, and every robust model absorbs it (Section 4.6).

## 2 Related work

**Truncation samplers.** Top- $k$  sampling draws from only the  $k$  most probable tokens; it was introduced for story generation, where likelihood-maximizing decoding produced generic and repetitive text (Fan et al., 2018). Holtzman et al. (2020)

showed that a fixed  $k$  fits poorly because the next-token distribution is sometimes peaked and sometimes flat, and proposed nucleus sampling, which keeps the smallest set of tokens whose probabilities sum to a target  $p$ . Min- $p$  keeps only the tokens whose probability is at least a set fraction of the top token's, so the cut is tight where the model is confident and loose where it is uncertain (Nguyen et al., 2025). Top- $n\sigma$  cuts in logit space instead, keeping tokens within  $n$  standard deviations of the top logit; temperature rescales all logits equally, so the surviving set does not change with temperature (Tang et al., 2024). Further refinements of the cut exist, among them typical sampling and epsilon and eta sampling (Meister et al., 2023; Hewitt et al., 2022). Welleck et al. (2020) proved the worst case of the failure all these rules target, showing that standard decoders can place probability on generations that never terminate. In contrast to the sampler papers, which compare truncation rules against each other, we test whether the failure they all guard against occurs at deployment temperatures at all.

**Reproducibility of sampler gains.** Schaeffer et al. (2025) reanalyzed min- $p$  on nine models and 31 temperatures, giving every sampler an equal number of hyperparameter settings, and found its reported gains "largely indistinguishable" from the other samplers'. A second variable in such a comparison is the order in which temperature and truncation are applied. The order matters because the truncation threshold is computed on whichever distribution the rule sees: applied before temperature, the rule cuts the model's original, sharp distribution; applied after, it cuts one that temperature has already flattened, so the same nominal setting keeps a different set of tokens. We report statistical uncertainty throughout and run the two chain orders as explicit arms (Section 3). That critique pooled its models together, because its question was about the sampler; ours is about the model, so we keep every model separate and ask whether the collapse happens on each one. Garces Arias et al. (2024) sweeps decoding hyperparameters on open-ended generation with quality and diversity metrics, while our tasks have one checkable answer. Shi et al. (2024) compare decoding methods on closed-ended tasks and report that the best method changes with the task and the model, and that quantization alters how robust a model is to the choice; the comparison is again among methods, and it leaves our premise question open. Song et al. (2025) documents per-

model gaps between greedy decoding and sampling on reasoning benchmarks and treats the gaps as evaluation noise to report; we treat the gap as the object of study and trace it to a termination failure in one model family.

**Robustness to decoding randomness.** Renze and Guven (2024) sampled temperatures from 0.0 to 1.6 on multiple-choice problem solving and found no statistically significant effect on accuracy between 0.0 and 1.0. Our results replicate that null inside the deployment band. But the null is pooled across models, and pooling hides the outlier. Our per-model ladder shows the heterogeneity beginning just past  $T=1.0$ , where Llama-3.1-8B falls while every other model holds (Section 4.1). The nearest study is Grover (2026), which also asks what makes a model robust to decoding randomness and concludes that instruction tuning is the primary determinant. Our data refine that conclusion. All seven of our models are instruction-tuned and one still collapses, so instruction tuning cannot be the whole answer. Four design differences let our grid observe what theirs cannot: their sweep stops at  $T=1.0$ , and the Llama collapse happens between 1.0 and 1.3; their robust models score 11 to 30% on synthetic tasks, so there is little accuracy left to lose; their instruct-versus-base contrast changes quantization at the same time (4-bit versus full precision), while we sweep quantization as its own axis; and they test no truncation samplers. Fastowski et al. (2025) connect the entropy of a model’s token distributions to its temperature sensitivity on factual question answering and report that larger models are more stable. Our logit probe supports the entropy link (Section 4.6). Size does not decide the outcome in our grid, because Qwen3-1.7B holds where the larger Llama-3.1-8B collapses.

### 3 Experimental setup

The study is a factorial grid over four axes: model, quantization level, decoding arm, and temperature, crossed with two tasks. Every claim in the paper is a comparison between cells of this grid, so everything outside the axes is pinned: one llama.cpp commit, each model’s official chat template rendered client-side, the native completion API, and per-record seeds derived deterministically from the cell identity. An accuracy difference between two cells is then attributable to the axes, up to sampling noise that the bootstrap below quantifies.

**Decoding.** Each cell runs eight arms: a deterministic greedy anchor, plain temperature sampling, and the four truncation rules of Section 2, each at the setting its source paper uses or recommends: top- $p$  0.95 (Holtzman et al., 2020), top- $k$  40 (Radford et al., 2019), min- $p$  0.05 (Nguyen et al., 2025), top- $n\sigma$  1.0 (Tang et al., 2024). Because the order of temperature and truncation changes which tokens survive (Section 2), we pin one order, temperature first, across the truncation arms, and add two ablation arms that instead apply temperature last (the llama.cpp default) for top- $p$  and min- $p$ . The other two rules need no ablation arm, because temperature scaling preserves the token ranking, so top- $k$  keeps the same set under either order, and top- $n\sigma$  keeps the same set at any temperature by construction (Section 2). Temperatures  $T \in \{0.7, 1.0, 1.3\}$  span the deployment band. The defaults that models and engines ship sit between 0.6 and 1.0 (Llama-3.1’s generation config sets 0.6, llama.cpp defaults to 0.8, the OpenAI and Anthropic APIs to 1.0),<sup>1</sup> so 0.7 and 1.0 bracket common practice and 1.3 sits just above it. Section 4.4 extends the axis to  $T=2.0$ . Every stochastic arm runs  $K=3$  repetitions per item; greedy decoding is deterministic, so  $K=1$ .

**Tasks.** We evaluate GSM8K, grade-school math word problems (Cobbe et al., 2021) (0-shot chain-of-thought with the prompt of Meta’s Llama evaluation suite (Meta AI, 2024c), numeric exact match), and MMLU-Pro, ten-option multiple-choice questions across 14 categories (Wang et al., 2024) (0-shot chain-of-thought, letter exact match),  $N=50$  items each, with parsers faithful to lm-eval-harness (Gao et al., 2023). Both tasks ask the model to generate a chain of reasoning and land on one exactly checkable answer, and the study needs both properties: exact-match grading keeps effects of a few points free of judge noise, and a derailment can only show up if there is a long generation to derail. The two tasks differ in generation length, with MMLU-Pro’s chains running roughly twice as long as GSM8K’s, a contrast Section 4.4 uses to test whether longer generations fail earlier. Both tasks grade a generation by parsing a final answer out of the generated text, so the parser decides how a generation that never reaches an answer is graded. We therefore log the parse path of every record (strict final-answer sentence, flexible fallback, or

<sup>1</sup>Llama-3.1-8B-Instruct’s generation\_config.json, llama.cpp’s default sampler parameters, and the OpenAI and Anthropic API references, accessed July 2026.

failure) and report the parse failure rate as a metric of its own; a failed parse grades as incorrect. Section 4.3 shows that this metric carries the main result. Every contrast in the study compares cells on the same frozen items.

**Models and quantization.** The grid holds seven instruction-tuned models from five model families: Llama-3.1-8B (Grattafiori et al., 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Qwen2.5-7B (Yang et al., 2024), Qwen3-1.7B/4B/8B (Yang et al., 2025) (thinking off; a thinking-on spot-check in Appendix G), and Gemma-3-12B (Gemma Team, 2025). A follow-up temperature-only cell adds two more Llama generations, Llama-3-8B (Meta AI, 2024a) and Llama-3.2-3B (Meta AI, 2024b) (Section 4.1). Each model runs at four quantization levels (Q8\_0, Q6\_K, Q4\_K\_M, Q3\_K\_M), all imatrix GGUF conversions from a single provider so the quantization axis is not confounded by the conversion recipe. A BF16 anchor on Qwen3-4B matches Q8 within noise, which lets Q8 stand in for full precision.

**Statistical inference.** All confidence intervals come from a nonparametric bootstrap ( $B=2000$ ) clustered on task items: we resample the 50 items rather than individual generations, because the  $K=3$  repetitions of one item share that item’s difficulty and treating them as independent would understate the uncertainty. Where a comparison shares items across its two sides, which is everywhere in the grid, the bootstrap is paired.

**Hardware.** Generation ran on RTX 2080 Ti (11 GB) nodes of a university Slurm cluster and one RTX 5070 Ti (16 GB) workstation, with per-architecture CUDA builds of the same pinned llama.cpp commit; the study totals about 204,000 graded generations. Full reproducibility details, including commit and file hashes, are in Appendix K.

## 4 Results

### 4.1 The collapse is model-bound

We begin with plain temperature sampling, no truncation, because whatever accuracy the intact tail loses as temperature rises is the failure the truncation samplers of Section 2 exist to prevent. Figure 1 plots the drop from  $T=0.7$  to  $T=1.3$ , pooled over quantization levels, and Table 1 gives the exact values; Section 4.5 shows quantization produces no collapse on any model.

The models do not split cleanly into robust and fragile. The Qwen3 family is a tight null, Qwen2.5,

Table 1: Plain-temperature accuracy drop  $T0.7 \rightarrow T1.3$  (pp, 95% item-clustered bootstrap CIs; pooled over quantization) and sampler max–min accuracy spread at  $T=1.3$  (pp; CIs in Figure 4). †: lineage cell, Q8-only and temperature-only; accuracy levels in Appendix D.

| Model           | Task     | drop [95% CI]      | spread |
|-----------------|----------|--------------------|--------|
| Llama-3.1-8B    | GSM8K    | +34.0 [28.5, 39.7] | 36.7   |
|                 | MMLU-Pro | +40.8 [31.8, 50.3] | 41.8   |
| Mistral-7B-v0.3 | GSM8K    | +3.5 [−1.2, 8.5]   | 5.3    |
|                 | MMLU-Pro | +8.0 [3.5, 13.2]   | 10.0   |
| Qwen2.5-7B      | GSM8K    | +4.0 [1.2, 7.3]    | 3.8    |
|                 | MMLU-Pro | +4.8 [1.3, 8.8]    | 5.7    |
| Gemma-3-12B     | GSM8K    | +2.0 [−0.0, 4.8]   | 3.3    |
|                 | MMLU-Pro | +3.7 [0.2, 7.7]    | 5.8    |
| Qwen3-8B        | GSM8K    | +0.5 [−1.5, 2.6]   | 1.8    |
|                 | MMLU-Pro | +1.0 [−2.4, 4.8]   | 3.5    |
| Qwen3-4B        | GSM8K    | +2.1 [0.4, 3.9]    | 1.3    |
|                 | MMLU-Pro | +0.7 [−1.0, 2.5]   | 1.7    |
| Qwen3-1.7B      | GSM8K    | −0.3 [−3.8, 3.0]   | 3.2    |
|                 | MMLU-Pro | +0.7 [−3.4, 5.0]   | 5.5    |
| Llama-3-8B†     | GSM8K    | +4.0 [−2.7, 11.3]  | —      |
|                 | MMLU-Pro | +16.7 [9.3, 24.0]  | —      |
| Llama-3.2-3B†   | GSM8K    | +38.0 [29.3, 47.3] | —      |
|                 | MMLU-Pro | +36.7 [26.7, 46.0] | —      |

Gemma, and Mistral drop 2 to 8 pp with at least one task CI excluding zero, and Llama drops 34 to 41 pp. Three controls rule out the obvious confounds. Accuracy headroom does not explain the collapse, because Qwen3-1.7B (74% on GSM8K) sits in Llama’s accuracy regime (78%) with the opposite temperature behavior. Model age does not explain it either, since Mistral-7B-v0.3 predates Llama-3.1 and is robust. The stratified MMLU-Pro replication rules out the item subset, giving Llama +43.3 pp [31.3, 55.3], Qwen2.5 +6.0, and Gemma +7.3.

A single collapsing checkpoint could still be a quirk of that release rather than a family trait, so a follow-up cell repeats the pure-temperature axis on two more Llama generations, Llama-3-8B and Llama-3.2-3B (Q8, same frozen items,  $K=3$ ; drops in Table 1, full accuracy levels in Appendix D). Llama-3.2-3B collapses like Llama-3.1-8B (+38.0 pp on GSM8K, +36.7 on MMLU-Pro) and reproduces the degeneration signature of Section 4.3: about half of all generations run to the token cap, and accuracy among well-formed answers stays at or above the greedy level. Llama-3-8B sits between the two: a null on GSM8K but +16.7 pp on MMLU-Pro, the longer task, a larger drop than any non-Llama model shows in Figure 1. Three Llama generations produce the three largest MMLU-Pro drops in the study, so the fragility is

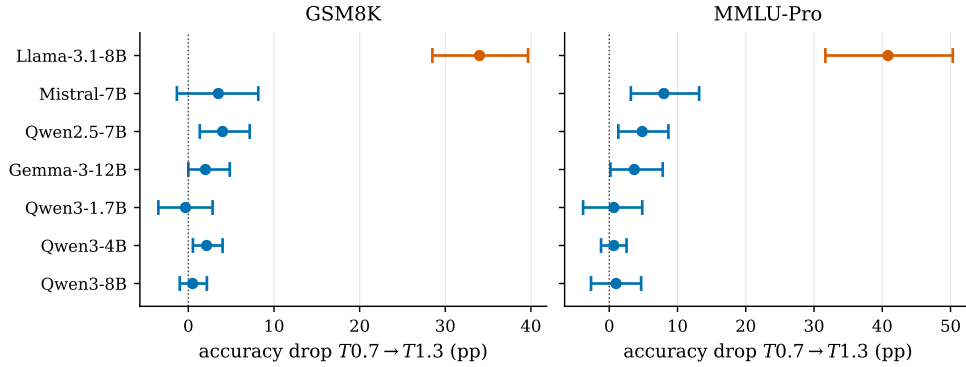


Figure 1: The pure-temperature collapse is model-bound: accuracy drop  $T_{0.7} \rightarrow T_{1.3}$ , points with 95% item-clustered bootstrap CIs.

a family trait; what changed in the family’s post-training is the open question of Section 5.

## 4.2 Sampler choice barely matters before the cliff

The previous subsection varied temperature with the sampler fixed; this one holds temperature at  $T=1.3$ , the top of the deployment band, and varies the sampler, the axis the truncation literature tunes. The maximum minus minimum accuracy across the seven stochastic sampler arms is 1.3 to 10.0 pp on every robust model (Table 1, spread column; per-arm accuracy grids in Appendix A), and even those numbers overstate the axis: seven equally good samplers would still show a positive spread, because the maximum picks the estimate that got lucky and the minimum the one that did not. The true sampler differences are smaller still. Llama’s spread measures something else. Its arms range from plain temperature, which has already collapsed at  $T=1.3$ , to top- $n\sigma$ , which truncates hard enough to hold (Section 4.4), so on Llama the spread records how much of the collapse each rule prevents.

Chain order (temperature first versus last) is a no-op on the robust models ( $\pm 1$ –2 pp). On Llama at  $T=1.3$ , temperature-last truncation is more protective (top- $p$  64%  $\rightarrow$  73% on GSM8K), as the threshold mechanics of Section 2 predict (both orders in Appendix F).

## 4.3 The collapse is degeneration, not wrong answers

An accuracy drop can come from two failures: completed answers that are wrong, or generations that never reach an answer. The parse path that Section 3 logs for every record separates the two, so

we decompose Llama’s collapse along it (Figure 2; full decomposition and example degenerate outputs in Appendix B).

Nearly all of the drop is the second failure. At  $T=1.3$ , 53.8% of Llama’s GSM8K generations and 89.2% of its MMLU-Pro generations derail into token-level noise and run to the token cap. Among the generations that do produce a well-formed final answer (a strict parse: the generation ends with the pinned final-answer sentence), Llama still scores 97.3% on GSM8K at  $T=1.3$ . That number needs one correction before it can be taken at face value. Survival is selective, because easier items produce shorter generations and shorter generations have less room to derail. The fair baseline is therefore greedy decoding restricted to the same surviving items, which scores 95.8% (against 76.0% over all items). Conditional on surviving, then, accuracy at  $T=1.3$  matches greedy decoding (97.3 vs. 95.8% on GSM8K; 76.9 vs. 78.8% on MMLU-Pro).

The two tasks disguise one failure mode differently. Degenerate text almost always contains a number, so the GSM8K flexible last-number fallback grades a derailed generation as a wrong answer while the MMLU-Pro letter parser reports a parse failure. The underlying event is the same, a generation that derails before it reaches an answer.

## 4.4 The cliff map: robustness is cliff location

Section 4.1 leaves open whether the robust models lack the failure or merely place it beyond the deployment band, so we extend the temperature axis to  $T \in \{1.3, 1.5, 1.7, 2.0\}$  on three models spanning the gradient of Figure 1: Llama-3.1-8B, Qwen2.5-7B, and Gemma-3-12B (Q8, both tasks,  $N=50$ ,  $K=3$ ). Figure 3 plots the accuracy trajectories, and Table 2 gives the exact drops against each

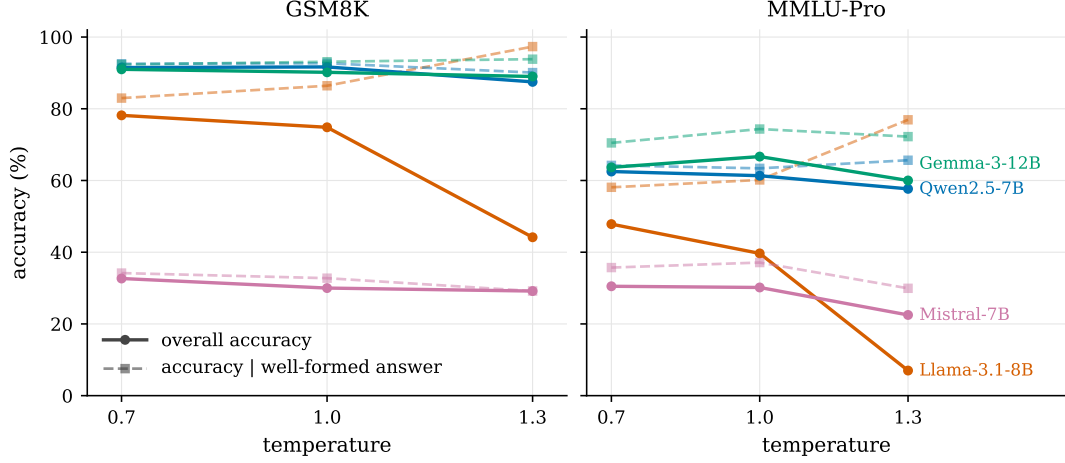


Figure 2: Overall accuracy (solid) versus accuracy among well-formed, strict-parsed answers (dashed) under plain temperature sampling; four models spanning the robustness gradient (the three Qwen3 models are flat and omitted for legibility). The dashed lines stay flat: among well-formed answers, accuracy barely moves with temperature.

model’s own  $T=0.7$  anchor. Every model turns out to have a cliff, and what distinguishes the models is where it sits. On GSM8K the three models collapse in their Figure 1 order, with cliffs roughly 0.2 apart in temperature. At  $T=1.5$  Llama’s drop is disjoint from the other two models’, and at  $T=1.7$  Qwen2.5’s is disjoint from Gemma’s. Within each model the cliff sits earlier on MMLU-Pro, the task with roughly twice the generation length, exactly what a compounding per-token derail risk predicts. There Llama has already collapsed at  $T=1.3$ , and Qwen2.5 and Gemma fall together at  $T=1.5$ .

Truncation shifts the cliff right in order of tail-cutting strength: top- $p$  buys about 0.2 in temperature, min- $p$  holds 70% at  $T=1.5$  on Llama GSM8K where plain temperature is at 9%, and top- $n\sigma$  is flat to  $T=2.0$  on all three models, as its temperature-invariant construction predicts. The strict-parse rate tracks accuracy almost one-to-one along the whole ladder, so the cliff *is* the degeneration cliff. This reconciles the sampler papers with our null result, because min- $p$  and top- $n\sigma$  are validated at temperatures of 1.5 to 3 (Nguyen et al., 2025; Tang et al., 2024), past every model’s deployment range, where their gains are real and ordered top- $n\sigma \gg$  min- $p \gg$  top- $p$  (the  $T=2-3$  probe in Appendix E).

#### 4.5 Quantization and answer agreement

The results so far pool over the four quantization levels. For that to be safe, quantization must not cost accuracy on its own, and it must not change what temperature and the samplers do. It costs little. From Q8\_0 down to Q3\_K\_M, greedy accu-

Table 2: The cliff ladder: accuracy drop against each model’s own  $T=0.7$  anchor under plain temperature (pp, 95% item-clustered bootstrap CIs; Q8).

| $T$             | Llama-3.1-8B       | Qwen2.5-7B         | Gemma-3-12B        |
|-----------------|--------------------|--------------------|--------------------|
| <i>GSM8K</i>    |                    |                    |                    |
| 1.3             | +32.7 [24.7, 41.3] | +4.0 [−0.7, 9.3]   | +2.7 [−2.0, 8.0]   |
| 1.5             | +68.0 [58.0, 76.7] | +24.0 [15.3, 33.3] | +14.7 [8.7, 21.3]  |
| 1.7             | +76.0 [66.7, 84.7] | +70.0 [62.0, 77.3] | +42.7 [34.7, 50.7] |
| 2.0             | +77.3 [67.3, 86.0] | +90.7 [82.7, 97.3] | +84.0 [76.7, 90.7] |
| <i>MMLU-Pro</i> |                    |                    |                    |
| 1.3             | +44.7 [34.0, 56.0] | +8.0 [0.0, 16.7]   | +3.3 [−3.3, 10.0]  |
| 1.5             | +46.0 [35.3, 57.3] | +40.0 [28.7, 52.0] | +39.3 [29.3, 48.7] |
| 1.7             | +48.0 [36.7, 60.0] | +61.3 [48.7, 74.0] | +57.3 [45.3, 68.0] |
| 2.0             | +48.0 [36.7, 59.3] | +65.3 [53.3, 78.0] | +63.3 [51.3, 75.3] |

racy is unchanged on every model of 7B parameters or more, and a real cost appears only on the two smallest models, 7 pp on Qwen3-4B and 9 pp on Qwen3-1.7B. It changes nothing about the collapse either. Recomputed separately at each level, the plain-temperature drop of Section 4.1 shows Llama-3.1 collapsing at every level, with drops of 30 to 47 pp across levels and tasks, while no robust model exceeds 12 pp at any single level. The BF16 anchor on Qwen3-4B shows the same null drop as its quantized versions, and Llama’s sampler ordering at  $T=1.3$  is identical at all four levels (per-level values in Appendix H).

A second check separates robustness from determinism, because flat accuracy across temperatures is also what a model would show if its next-token distributions were peaked enough that sampling returns the greedy path every time. The  $K=3$  repetitions decide between the two readings. Each item

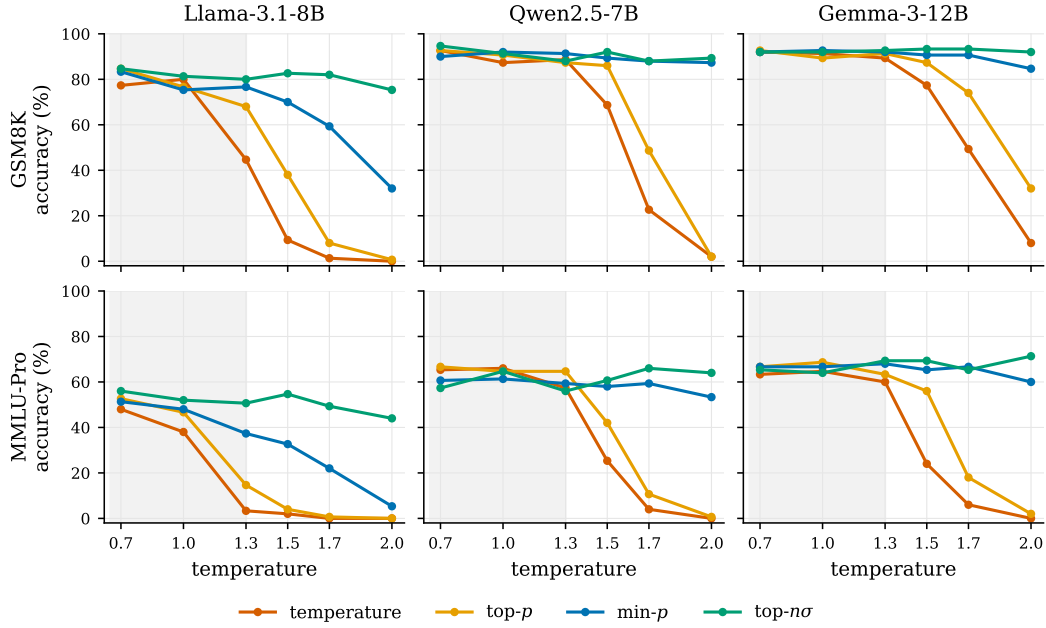


Figure 3: The cliff map. Accuracy versus temperature per sampler (the  $T \leq 1.0$  points come from the main matrix; all at Q8). Every model has a cliff; where it sits is the model property. Truncation shifts the cliff right by tail-cutting strength.

carries three independently sampled generations under identical settings, and if the robust models were merely deterministic, the three would be near copies of one another. Instead, the token sequences differ across repetitions, and on the robust models the answers still converge: 1.1 to 1.4 distinct parsed answers per item on GSM8K at  $T=1.3$ , moved by at most  $+0.17$  when temperature rises from 0.7 to 1.3. These models sample genuinely different token sequences and land on the same answer. Llama-3.1 rises from 1.5 to 2.3 under the same change, as expected when each repetition derails independently (full table, and Mistral’s floor-driven scatter, in Appendix I).

#### 4.6 Toward a mechanism: per-token uncertainty ranks with fragility

If the collapse is a derailment process, fragile models should carry more per-token uncertainty before any derailment happens. To measure that uncertainty we decode the 100 frozen prompts (the 50 items of each task) greedily on every model at Q8 and record the raw top-20 next-token probabilities at each position. Each prompt is summarized by four statistics of these distributions: an entropy lower bound (only the top 20 probabilities are observed), the top-1 probability and its margin over the top-2, and the fraction of positions where the top-1 probability falls below 0.5. We decode greed-

ily so that every model is measured on its own modal path; the statistics then describe the undisturbed trajectory, unaffected by any sampled derailment.

Mean greedy-path entropy ranks the seven models’ temperature drops at Spearman  $\rho = +0.96$  (flat-position fraction  $+0.96$ ; top-1 margin  $-0.96$ ; plotted in Appendix J). The flatter a model’s next-token distributions on its own greedy path, the more accuracy it loses to temperature. The summaries cannot separate one pair, though. Llama-3.1 and Mistral are nearly tied on every statistic we record, 0.33 versus 0.32 nats of entropy and  $1.0 \times 10^{-3}$  probability mass beyond the top 20 for both, yet Llama drops 40.8 pp on MMLU-Pro and Mistral 8.0. Uncertainty on the undisturbed path predicts the ordering of the drops and leaves their size unexplained.

Generation lengths rule out one candidate mechanism, derail-and-recover. A recovered detour leaves extra tokens behind, yet on every robust model the length distribution of terminated generations is essentially unchanged from  $T=0.7$  to  $T=1.3$  (90th-percentile ratio 0.85 to 1.19), and Llama’s survivors at  $T=1.3$  are *shorter*, the survivorship of Section 4.3 at work. A generation that derails does not come back, and its per-token risk compounds.

That per-token risk can be estimated from the

probe itself. From the recorded top-20 arrays we compute, for each position, the probability that sampling at  $T=1.3$  picks a token from beyond the model’s own top 20; we call this the escape hazard, the per-token probability of a genuine tail draw. Llama’s is 0.0033 per position and Mistral’s 0.0034, about 0.85 expected tail draws per 256 positions for both, and across the seven models the hazard ranks the drops at the same  $\rho = +0.96$ . The two models therefore take their first off-distribution token equally often; what separates them must be what happens after that token has entered the model’s own context.

A perturbation experiment measures that directly. For each model and frozen prompt we greedily-decode a 48-token prefix, replace the next token with the model’s own rank-20 candidate at that position (median raw probability  $2 \times 10^{-9}$ , a genuine tail draw), and continue greedily for 256 tokens, alongside a control continuation of the unedited prefix. Judged by the text alone, the forced token is absorbed. The continuations stay locally coherent, and Llama’s reach the token cap no more often than its controls (14.0% for both on GSM8K). The instability appears in the escape hazard over the 32 positions after the injection: Llama’s rises to 2.3 times its baseline (+0.0042 per position), Mistral’s to 1.35 times (+0.0012), and no robust model gains more than +0.0015. This per-model destabilization ranks the temperature drops at  $\rho = +0.86$  (the entropy increase, +0.82). It is the compounding loop anticipated in the introduction, measured directly. An off-distribution token in context raises the probability of drawing the next one, and the size of that rise is what distinguishes the fragile models (per-model probe numbers in Appendix J). With seven models these are consistency checks, not a confirmed causal account, and we report them as such.

## 5 Discussion

**For practitioners.** At deployment temperatures the consequential choice is the model. On a robust model every configuration we test stays within a few points of every other, so time spent tuning the sampler buys little; on a fragile model the same sweep spans up to 42 points. Screening for the fragility is cheap, because the collapse is large and needs no judge to detect. Fifty items under plain temperature at the top of the deployment band suffice, and a fragile model reveals itself in the strict-

parse rate alone. In production, temperature breaks termination and leaves answer selection intact, so a derailed generation carries no well-formed final answer and simple output validation catches it, while the rate of silent errors, well-formed but wrong answers, barely moves.

**For the sampler literature.** Truncation samplers solve a real problem, and the problem sits above the temperatures at which models are deployed, because below the cliff there is no failure left for a sampler to prevent (Section 4.4). A sampler evaluation should therefore say where its models’ cliffs sit, since a gain measured past one model’s cliff need not exist on a model whose cliff lies further out.

## 6 Conclusion

Across nine models from five families, the choice of truncation sampler is a second-order effect at deployment temperatures. Robust models hold within a few points under every sampler we test, while the Llama family loses up to 41 points under plain temperature across three of its generations. The collapse arrives in every model at some temperature above its deployment range, robustness is how far above, and truncation samplers push that temperature higher without mattering below it. These results argue for screening temperature fragility before deployment rather than tuning the sampler after it. The data also fix the signature a mechanism study must explain: termination failure with intact answer selection, off-distribution tokens arriving at the same per-token rate in every model, and a post-injection destabilization only the fragile family shows. The open question our data sharpen but cannot answer is what, in post-training, moves the cliff.

## Limitations

Our sample is 50 items per task with  $K=3$  repetitions per cell; the small item count is what makes a factorial of about 204,000 generations feasible. Item-clustered bootstrap CIs accompany every estimate, and the headline effects are 5 to 10 times their CI widths. Mistral-7B sits near floor on GSM8K (32.7% at  $T=0.7$ ), which limits collapse-detection power in that cell; its MMLU-Pro drop is measured away from floor and its CI excludes zero. The frozen MMLU-Pro subset is single-category (business) because the test split is sorted by category; the stratified  $N=50$  replication of Appendix C re-

produces the headline result.

The scope is open-weight instruction-tuned models of 1.7B to 12B parameters, served through one inference engine. Whether larger or proprietary models share the Llama family’s fragility is untested. Both tasks are English, zero-shot, and objective: a single exactly checkable answer is what lets the parser measure degeneration. The original use case of truncation samplers, open-ended generation with no checkable answer, lies outside this design, so our null result on sampler gains does not extend to it. Quantization covers llama.cpp k-quants only, one recipe by design so that the quantization axis varies precision and nothing else; other recipes, and other engines’ sampler implementations, may behave differently. Thinking mode is off throughout the main grid; the thinking-on spot-check of Appendix G agrees with it.

Every fragile model we observe is a Llama. The lineage cell of Section 4.1 shows the fragility spans three generations, so it is a family trait rather than a quirk of one checkpoint, but with a single fragile family we cannot say how common such families are.

Three analysis caveats. The max-minus-min sampler spread is biased upward, so it overstates the differences between samplers; the bias is conservative for our claim that those differences are small. Accuracy conditional on a strict parse carries item survivorship, which Section 4.3 quantifies and normalizes. The mechanism probes correlate seven models, and Section 4.6 presents them as consistency checks rather than a causal account.

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## A Full sampler–temperature grids

Figure 4 plots the per-model sampler max–min spread of Section 4.2 with its confidence intervals. Tables 3 and 4 give the full main-matrix grids: accuracy for every sampler arm and temperature, per model, pooled over quantization levels.

## B Degeneration decomposition

Table 6 decomposes accuracy under plain temperature sampling by parse path: the strict final-answer-sentence rate, the parse-failure rate, the fraction of generations that ran to the token cap (including server-recovered degenerate outputs), and accuracy conditional on a strict parse. Figure 5 shows two

representative degenerate outputs. The pattern is consistent: a coherent opening, a mid-generation derailment, then token noise until the cap. The model never reaches an answer at all.

Table 5 gives the length-censoring evidence used in Section 4.6: run-to-cap rates and the 90th percentile of completion length among *terminated* generations at  $T=0.7$  versus  $T=1.3$ . Robust models are unchanged (ratio 0.85 to 1.19); Llama’s surviving generations shorten ( $\times 0.73$  GSM8K,  $\times 0.55$  MMLU-Pro), so derailment is absorbing and survival is decided by whether the generation finishes before the first bad draw.

## C Stratified MMLU-Pro replication

The frozen  $N=50$  MMLU-Pro head subset is single-category (business) because the test split is sorted by category. Table 7 repeats the pure-temperature axis (Q8,  $K=3$ ) on a deterministic stratified subset with 3–4 items from each of the 14 categories. The collapse ordering reproduces.

## D Llama lineage cell

The lineage cell repeats the pure-temperature axis (greedy anchor plus plain temperature at  $T \in \{0.7, 1.0, 1.3\}$ ) on Llama-3-8B and Llama-3.2-3B: Q8\_0, both frozen  $N=50$  tasks,  $K=3$ , 2,000 generations, same pipeline, prompts, and seed derivation as the main matrix. Table 8 reports both models next to the Q8-only Llama-3.1-8B slice of the main matrix, which is why the Llama-3.1 numbers differ slightly from the quantization-pooled Table 1. Greedy anchors are ordinary (Llama-3-8B 76.0% GSM8K / 40.0% MMLU-Pro; Llama-3.2-3B 74.0 / 42.0%). Llama-3.2-3B reproduces the full degeneration signature at  $T=1.3$ : strict-parse rate  $98.7 \rightarrow 41.3\%$  on GSM8K and  $82.7 \rightarrow 6.0\%$  on MMLU-Pro, 50 to 51% of generations at the token cap, and accuracy conditional on a strict parse at or above the greedy level (85.5 and 55.6%). Llama-3-8B’s MMLU-Pro drop has a milder surface: strict-parse  $97.3 \rightarrow 65.3\%$  with only 4.7% at the cap, so its lost generations mostly terminate as parse failures short of the cap.

## E High-temperature regime

Table 9 reports a small probe ( $N=15$ ,  $K=3$ , Q8) in the regime where min- $p$  and top- $n\sigma$  report their largest gains. The sampler ordering is top- $n\sigma \gg$  min- $p \gg$  top- $p \approx$  temperature; the zero cells are

Table 3: GSM8K accuracy (%) by sampler and temperature, pooled over quantization.

| Model        | Sampler              | $T=0.7$ | $T=1.0$ | $T=1.3$ |
|--------------|----------------------|---------|---------|---------|
| Llama-3.1-8B | greedy ( $T=0$ )     | 78.0    | –       | –       |
|              | temperature          | 78.2    | 74.8    | 44.2    |
|              | top- $k$             | 79.8    | 75.2    | 58.7    |
|              | top- $p$             | 81.5    | 78.0    | 63.8    |
|              | min- $p$             | 81.2    | 78.0    | 72.0    |
|              | top- $n\sigma$       | 78.8    | 80.7    | 80.8    |
|              | top- $p$ (temp-last) | 80.8    | 78.7    | 73.0    |
|              | min- $p$ (temp-last) | 78.7    | 81.3    | 76.2    |
| Mistral-7B   | greedy ( $T=0$ )     | 31.0    | –       | –       |
|              | temperature          | 32.7    | 30.0    | 29.2    |
|              | top- $k$             | 31.8    | 31.7    | 29.8    |
|              | top- $p$             | 28.2    | 31.0    | 34.2    |
|              | min- $p$             | 32.3    | 28.3    | 28.8    |
|              | top- $n\sigma$       | 29.5    | 30.7    | 30.2    |
|              | top- $p$ (temp-last) | 29.5    | 31.2    | 29.5    |
|              | min- $p$ (temp-last) | 29.5    | 32.0    | 30.5    |
| Qwen2.5-7B   | greedy ( $T=0$ )     | 93.5    | –       | –       |
|              | temperature          | 91.5    | 91.7    | 87.5    |
|              | top- $k$             | 90.2    | 88.5    | 88.7    |
|              | top- $p$             | 92.2    | 91.2    | 90.3    |
|              | min- $p$             | 90.8    | 90.5    | 91.0    |
|              | top- $n\sigma$       | 92.7    | 91.2    | 90.7    |
|              | top- $p$ (temp-last) | 90.8    | 91.3    | 90.3    |
|              | min- $p$ (temp-last) | 92.0    | 90.7    | 91.3    |
| Gemma-3-12B  | greedy ( $T=0$ )     | 92.0    | –       | –       |
|              | temperature          | 91.0    | 90.2    | 89.0    |
|              | top- $k$             | 91.8    | 91.7    | 90.7    |
|              | top- $p$             | 91.5    | 90.3    | 90.7    |
|              | min- $p$             | 91.5    | 91.3    | 92.3    |
|              | top- $n\sigma$       | 91.3    | 91.5    | 91.0    |
|              | top- $p$ (temp-last) | 90.2    | 91.8    | 91.5    |
|              | min- $p$ (temp-last) | 91.2    | 90.8    | 90.3    |
| Qwen3-1.7B   | greedy ( $T=0$ )     | 75.5    | –       | –       |
|              | temperature          | 73.3    | 73.7    | 73.7    |
|              | top- $k$             | 72.8    | 72.7    | 73.3    |
|              | top- $p$             | 73.8    | 75.0    | 72.7    |
|              | min- $p$             | 72.2    | 71.7    | 74.0    |
|              | top- $n\sigma$       | 74.3    | 73.5    | 72.3    |
|              | top- $p$ (temp-last) | 73.0    | 72.0    | 71.3    |
|              | min- $p$ (temp-last) | 74.7    | 73.3    | 74.5    |
| Qwen3-4B     | greedy ( $T=0$ )     | 95.2    | –       | –       |
|              | temperature          | 95.2    | 94.0    | 93.1    |
|              | top- $k$             | 94.0    | 94.5    | 94.1    |
|              | top- $p$             | 95.5    | 95.5    | 93.5    |
|              | min- $p$             | 95.1    | 94.7    | 92.9    |
|              | top- $n\sigma$       | 94.8    | 93.9    | 93.9    |
|              | top- $p$ (temp-last) | 94.8    | 95.2    | 94.3    |
|              | min- $p$ (temp-last) | 94.5    | 93.9    | 93.9    |
| Qwen3-8B     | greedy ( $T=0$ )     | 94.5    | –       | –       |
|              | temperature          | 95.0    | 94.2    | 94.5    |
|              | top- $k$             | 94.8    | 94.8    | 93.5    |
|              | top- $p$             | 94.0    | 94.3    | 93.7    |
|              | min- $p$             | 94.8    | 95.3    | 94.7    |
|              | top- $n\sigma$       | 96.3    | 94.2    | 94.3    |
|              | top- $p$ (temp-last) | 93.8    | 94.0    | 94.0    |
|              | min- $p$ (temp-last) | 96.2    | 94.7    | 95.3    |

Table 4: MMLU-Pro accuracy (%) by sampler and temperature, pooled over quantization.

| Model        | Sampler              | $T=0.7$ | $T=1.0$ | $T=1.3$ |
|--------------|----------------------|---------|---------|---------|
| Llama-3.1-8B | greedy ( $T=0$ )     | 54.0    | –       | –       |
|              | temperature          | 47.8    | 39.7    | 7.0     |
|              | top- $k$             | 51.2    | 43.3    | 26.8    |
|              | top- $p$             | 51.3    | 46.0    | 16.0    |
|              | min- $p$             | 51.2    | 50.8    | 39.5    |
|              | top- $n\sigma$       | 51.0    | 50.5    | 48.8    |
|              | top- $p$ (temp-last) | 50.3    | 46.7    | 37.5    |
|              | min- $p$ (temp-last) | 51.2    | 50.2    | 43.5    |
| Mistral-7B   | greedy ( $T=0$ )     | 35.5    | –       | –       |
|              | temperature          | 30.5    | 30.2    | 22.5    |
|              | top- $k$             | 32.5    | 32.3    | 28.3    |
|              | top- $p$             | 30.2    | 29.5    | 25.3    |
|              | min- $p$             | 33.8    | 30.7    | 31.0    |
|              | top- $n\sigma$       | 32.3    | 31.5    | 31.5    |
|              | top- $p$ (temp-last) | 29.2    | 30.2    | 32.5    |
|              | min- $p$ (temp-last) | 33.2    | 33.3    | 28.7    |
| Qwen2.5-7B   | greedy ( $T=0$ )     | 64.0    | –       | –       |
|              | temperature          | 62.5    | 61.3    | 57.7    |
|              | top- $k$             | 62.5    | 61.7    | 57.0    |
|              | top- $p$             | 65.5    | 62.8    | 62.7    |
|              | min- $p$             | 61.3    | 63.5    | 59.7    |
|              | top- $n\sigma$       | 62.0    | 63.5    | 61.0    |
|              | top- $p$ (temp-last) | 63.7    | 62.7    | 61.3    |
|              | min- $p$ (temp-last) | 64.2    | 63.0    | 62.3    |
| Gemma-3-12B  | greedy ( $T=0$ )     | 61.0    | –       | –       |
|              | temperature          | 63.7    | 66.7    | 60.0    |
|              | top- $k$             | 64.3    | 62.8    | 63.7    |
|              | top- $p$             | 65.7    | 66.7    | 63.8    |
|              | min- $p$             | 65.3    | 67.0    | 65.5    |
|              | top- $n\sigma$       | 65.2    | 65.8    | 65.8    |
|              | top- $p$ (temp-last) | 65.8    | 65.0    | 65.2    |
|              | min- $p$ (temp-last) | 65.3    | 63.7    | 63.8    |
| Qwen3-1.7B   | greedy ( $T=0$ )     | 45.5    | –       | –       |
|              | temperature          | 46.0    | 45.7    | 45.3    |
|              | top- $k$             | 47.5    | 46.5    | 45.0    |
|              | top- $p$             | 47.2    | 47.0    | 47.2    |
|              | min- $p$             | 47.0    | 47.7    | 46.2    |
|              | top- $n\sigma$       | 46.8    | 47.2    | 46.2    |
|              | top- $p$ (temp-last) | 47.7    | 46.8    | 43.5    |
|              | min- $p$ (temp-last) | 45.7    | 46.2    | 49.0    |
| Qwen3-4B     | greedy ( $T=0$ )     | 61.6    | –       | –       |
|              | temperature          | 61.9    | 62.5    | 61.2    |
|              | top- $k$             | 63.5    | 61.1    | 61.2    |
|              | top- $p$             | 62.8    | 61.3    | 62.8    |
|              | min- $p$             | 60.9    | 62.7    | 62.9    |
|              | top- $n\sigma$       | 62.4    | 62.4    | 61.5    |
|              | top- $p$ (temp-last) | 62.9    | 62.7    | 61.7    |
|              | min- $p$ (temp-last) | 61.3    | 63.6    | 61.5    |
| Qwen3-8B     | greedy ( $T=0$ )     | 60.5    | –       | –       |
|              | temperature          | 60.5    | 59.0    | 59.5    |
|              | top- $k$             | 60.2    | 61.3    | 57.0    |
|              | top- $p$             | 60.5    | 61.3    | 60.5    |
|              | min- $p$             | 61.0    | 59.2    | 59.7    |
|              | top- $n\sigma$       | 60.0    | 58.8    | 58.7    |
|              | top- $p$ (temp-last) | 60.7    | 59.5    | 60.5    |
|              | min- $p$ (temp-last) | 61.3    | 60.7    | 59.3    |

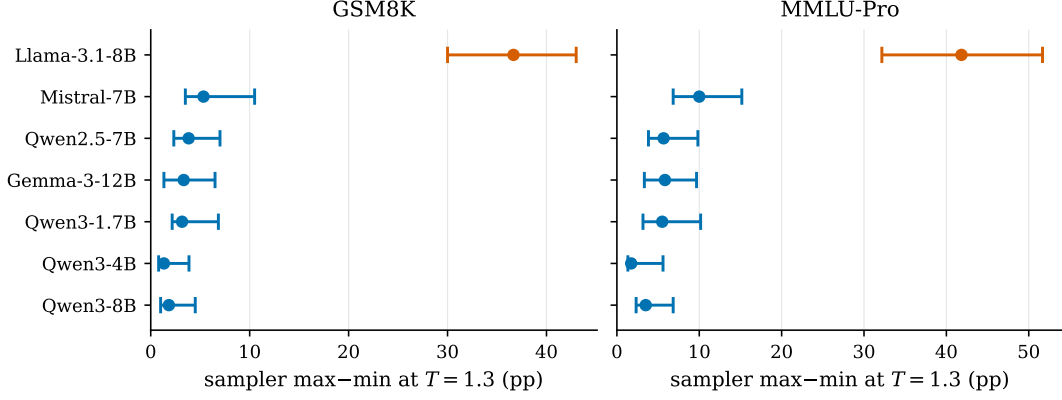


Figure 4: Sampler max-min accuracy spread at  $T=1.3$  per model (95% CIs).

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Figure 5: Two representative degenerate outputs (Llama-3.1-8B, Q8, plain temperature,  $T=1.3$ , MMLU-Pro; both ran to the 1024-token cap).

Table 5: Terminated-length censoring under plain temperature at  $T=0.7$  versus  $T=1.3$  (pooled over quantization). “At cap” includes server-recovered degenerate streams.

| Model        | Task     | at cap (%) |      | p90 length |     | ratio |
|--------------|----------|------------|------|------------|-----|-------|
|              |          | 0.7        | 1.3  | 0.7        | 1.3 |       |
| Llama-3.1-8B | GSM8K    | 5.2        | 53.8 | 318        | 232 | 0.73× |
|              | MMLU-Pro | 11.0       | 89.2 | 633        | 345 | 0.55× |
| Mistral-7B   | GSM8K    | 3.5        | 8.3  | 367        | 378 | 1.03× |
|              | MMLU-Pro | 1.0        | 12.5 | 628        | 750 | 1.19× |
| Qwen2.5-7B   | GSM8K    | 0.2        | 1.7  | 341        | 361 | 1.06× |
|              | MMLU-Pro | 2.5        | 10.2 | 771        | 732 | 0.95× |
| Gemma-3-12B  | GSM8K    | 2.0        | 5.3  | 363        | 344 | 0.95× |
|              | MMLU-Pro | 9.8        | 17.2 | 753        | 763 | 1.01× |
| Qwen3-1.7B   | GSM8K    | 2.0        | 2.2  | 324        | 339 | 1.05× |
|              | MMLU-Pro | 15.8       | 16.3 | 779        | 806 | 1.03× |
| Qwen3-4B     | GSM8K    | 0.9        | 0.8  | 353        | 361 | 1.02× |
|              | MMLU-Pro | 11.9       | 12.4 | 778        | 731 | 0.94× |
| Qwen3-8B     | GSM8K    | 2.3        | 2.3  | 379        | 389 | 1.03× |
|              | MMLU-Pro | 12.3       | 16.0 | 850        | 724 | 0.85× |

entirely degeneration-to-cap, and even the most robust model collapses under plain temperature.

## F Temperature-first versus temperature-last

Table 10 gives both chain orders for the two probability-space truncators. The two orders are indistinguishable on the robust models. On Llama at  $T=1.3$ , truncating before temperature scaling is more protective, worth +9.2 pp on GSM8K and +21.5 pp on MMLU-Pro for top- $p$  (+4.2 and +4.0 for min- $p$ ); the results section explains why.

## G Thinking-mode spot-check

The matrix runs Qwen3 with thinking off. A confirmatory spot-check (Qwen3-8B, Q8, thinking on, 450 generations,  $N=25$ ,  $K=2$ ) shows the same temperature robustness: under plain temperature,

GSM8K 94% at  $T=0.7$  versus 96% at  $T=1.3$ , and MMLU-Pro 56% versus 46% (small sample), with min- $p$  flat on both tasks. The robustness is not an artifact of suppressing the reasoning mode.

## H Per-quantization-level temperature drops

Figure 6 plots the direct cost of quantization on greedy accuracy. Table 11 recomputes the plain-temperature drop of Table 1 separately at each quantization level. Each cell pools 150 generations ( $N=50$ ,  $K=3$ ), so single-level estimates are noisier than the pooled ones. The BF16 anchor on Qwen3-4B shows the same null drop as its quantized versions (−1.3 pp GSM8K, −2.0 pp MMLU-

Table 6: Decomposition of the temperature effect (plain temperature, pooled over quantization; %).

| Model        | Task     | $T$ | acc  | strict | failed | at cap | acc strict |
|--------------|----------|-----|------|--------|--------|--------|------------|
| Llama-3.1-8B | GSM8K    | 0.7 | 78.2 | 93.0   | 0.0    | 5.2    | 83.0       |
|              | GSM8K    | 1.0 | 74.8 | 84.7   | 0.0    | 7.8    | 86.4       |
|              | GSM8K    | 1.3 | 44.2 | 43.8   | 0.5    | 53.8   | 97.3       |
|              | MMLU-Pro | 0.7 | 47.8 | 82.3   | 15.3   | 11.0   | 58.1       |
|              | MMLU-Pro | 1.0 | 39.7 | 64.8   | 29.0   | 23.7   | 60.2       |
|              | MMLU-Pro | 1.3 | 7.0  | 8.7    | 84.5   | 89.2   | 76.9       |
| Mistral-7B   | GSM8K    | 0.7 | 32.7 | 72.2   | 0.3    | 3.5    | 34.2       |
|              | GSM8K    | 1.0 | 30.0 | 67.7   | 0.0    | 4.3    | 32.8       |
|              | GSM8K    | 1.3 | 29.2 | 62.8   | 0.0    | 8.3    | 29.2       |
|              | MMLU-Pro | 0.7 | 30.5 | 84.0   | 12.8   | 1.0    | 35.7       |
|              | MMLU-Pro | 1.0 | 30.2 | 80.8   | 17.0   | 1.0    | 37.1       |
|              | MMLU-Pro | 1.3 | 22.5 | 73.5   | 21.2   | 12.5   | 29.9       |
| Qwen2.5-7B   | GSM8K    | 0.7 | 91.5 | 98.3   | 0.0    | 0.2    | 92.5       |
|              | GSM8K    | 1.0 | 91.7 | 95.7   | 0.0    | 0.5    | 92.7       |
|              | GSM8K    | 1.3 | 87.5 | 92.5   | 0.0    | 1.7    | 90.1       |
|              | MMLU-Pro | 0.7 | 62.5 | 97.3   | 2.3    | 2.5    | 64.2       |
|              | MMLU-Pro | 1.0 | 61.3 | 96.5   | 2.5    | 3.0    | 63.4       |
|              | MMLU-Pro | 1.3 | 57.7 | 87.3   | 10.5   | 10.2   | 65.6       |
| Gemma-3-12B  | GSM8K    | 0.7 | 91.0 | 98.0   | 0.0    | 2.0    | 92.3       |
|              | GSM8K    | 1.0 | 90.2 | 96.8   | 0.0    | 3.3    | 93.1       |
|              | GSM8K    | 1.3 | 89.0 | 94.5   | 0.0    | 5.3    | 93.8       |
|              | MMLU-Pro | 0.7 | 63.7 | 90.3   | 9.0    | 9.8    | 70.5       |
|              | MMLU-Pro | 1.0 | 66.7 | 89.7   | 8.2    | 10.3   | 74.3       |
|              | MMLU-Pro | 1.3 | 60.0 | 82.8   | 14.5   | 17.2   | 72.2       |
| Qwen3-1.7B   | GSM8K    | 0.7 | 73.3 | 20.3   | 0.0    | 2.0    | 73.8       |
|              | GSM8K    | 1.0 | 73.7 | 20.0   | 0.0    | 0.7    | 71.7       |
|              | GSM8K    | 1.3 | 73.7 | 19.0   | 0.0    | 2.2    | 71.1       |
|              | MMLU-Pro | 0.7 | 46.0 | 83.7   | 13.3   | 15.8   | 54.8       |
|              | MMLU-Pro | 1.0 | 45.7 | 82.2   | 15.5   | 16.3   | 55.4       |
|              | MMLU-Pro | 1.3 | 45.3 | 83.2   | 13.5   | 16.3   | 54.5       |
| Qwen3-4B     | GSM8K    | 0.7 | 95.2 | 94.3   | 0.0    | 0.9    | 95.3       |
|              | GSM8K    | 1.0 | 94.0 | 92.3   | 0.0    | 0.8    | 94.8       |
|              | GSM8K    | 1.3 | 93.1 | 95.6   | 0.0    | 0.8    | 93.7       |
|              | MMLU-Pro | 0.7 | 61.9 | 87.2   | 12.5   | 11.9   | 70.9       |
|              | MMLU-Pro | 1.0 | 62.5 | 87.3   | 12.0   | 10.8   | 71.6       |
|              | MMLU-Pro | 1.3 | 61.2 | 87.5   | 11.1   | 12.4   | 70.0       |
| Qwen3-8B     | GSM8K    | 0.7 | 95.0 | 48.2   | 0.0    | 2.3    | 98.3       |
|              | GSM8K    | 1.0 | 94.2 | 49.2   | 0.0    | 2.3    | 98.6       |
|              | GSM8K    | 1.3 | 94.5 | 52.2   | 0.0    | 2.3    | 98.1       |
|              | MMLU-Pro | 0.7 | 60.5 | 82.5   | 15.2   | 12.3   | 72.3       |
|              | MMLU-Pro | 1.0 | 59.0 | 80.5   | 15.5   | 14.8   | 72.3       |
|              | MMLU-Pro | 1.3 | 59.5 | 79.8   | 16.3   | 16.0   | 73.5       |

Table 7: Stratified versus head-of-split (business) MMLU-Pro subset; plain temperature accuracy (%) at Q8\_0.

| Model        | Subset     | $T$  |      |      |          |
|--------------|------------|------|------|------|----------|
|              |            | 0.7  | 1.0  | 1.3  | drop     |
| Llama-3.1-8B | stratified | 44.7 | 38.7 | 1.3  | +43.3 pp |
| Llama-3.1-8B | head-50    | 48.0 | 38.0 | 10.0 | +38.0 pp |
| Qwen2.5-7B   | stratified | 54.7 | 53.3 | 48.7 | +6.0 pp  |
| Qwen2.5-7B   | head-50    | 65.3 | 66.0 | 58.0 | +7.3 pp  |
| Gemma-3-12B  | stratified | 64.0 | 62.7 | 56.7 | +7.3 pp  |
| Gemma-3-12B  | head-50    | 63.3 | 64.7 | 60.0 | +3.3 pp  |

Pro).

## I Answer agreement across repetitions

Mean distinct parsed answers per item across the  $K=3$  repetitions under plain temperature (Table 12). Robust models sample diverse token sequences that converge on the same answer; a parse failure counts as one distinct answer, so the statistic is a divergence proxy; it is the per-item agreement that self-consistency decoding (Wang et al., 2023) exploits by majority vote. The parse-failure convention censors Llama-3.1 on MMLU-Pro at  $T=1.3$ : with 84.5% of generations failing to parse, the rep-

Table 8: Llama lineage: plain-temperature accuracy and  $T_{0.7} \rightarrow 1.3$  drop (pp, 95% item-clustered bootstrap CIs), all at Q8\_0. The Llama-3.1-8B rows are the Q8-only slice of the main matrix.

| Model        | Task     | acc (%) at $T$ |      | drop [95% CI]      |
|--------------|----------|----------------|------|--------------------|
|              |          | 0.7            | 1.3  |                    |
| Llama-3-8B   | GSM8K    | 75.3           | 71.3 | +4.0 [−2.7, 11.3]  |
|              | MMLU-Pro | 44.0           | 27.3 | +16.7 [9.3, 24.0]  |
| Llama-3.1-8B | GSM8K    | 77.3           | 47.3 | +30.0 [21.3, 38.7] |
|              | MMLU-Pro | 48.0           | 10.0 | +38.0 [28.0, 48.7] |
| Llama-3.2-3B | GSM8K    | 74.0           | 36.0 | +38.0 [29.3, 47.3] |
|              | MMLU-Pro | 40.0           | 3.3  | +36.7 [26.7, 46.0] |

Table 9: Accuracy (%) at  $T=2$  and  $T=3$  (45 observations per cell).

| Task     | Sampler        | Llama-3.1-8B |       | Gemma-3-12B |       |
|----------|----------------|--------------|-------|-------------|-------|
|          |                | $T=2$        | $T=3$ | $T=2$       | $T=3$ |
| GSM8K    | temperature    | 0            | 0     | 16          | 0     |
|          | top- $p$       | 0            | 0     | 27          | 0     |
|          | min- $p$       | 33           | 0     | 78          | 24    |
|          | top- $n\sigma$ | 71           | 80    | 87          | 89    |
| MMLU-Pro | temperature    | 0            | 0     | 0           | 0     |
|          | top- $p$       | 0            | 0     | 0           | 0     |
|          | min- $p$       | 0            | 0     | 58          | 0     |
|          | top- $n\sigma$ | 47           | 51    | 69          | 62    |

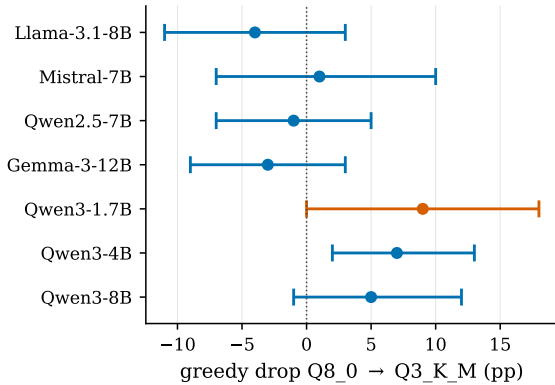


Figure 6: Greedy accuracy drop Q8\_0 → Q3\_K\_M per model (95% CIs), pooled tasks.

etitions agree on having no answer, and the statistic falls to 1.28. Mistral sits near 2.2 distinct answers at both temperatures, and its scatter has a different source: at a greedy GSM8K accuracy of 31%, most repetitions are wrong, and wrong repetitions rarely agree on the same wrong number at any temperature. The shift attributable to temperature is +0.27.

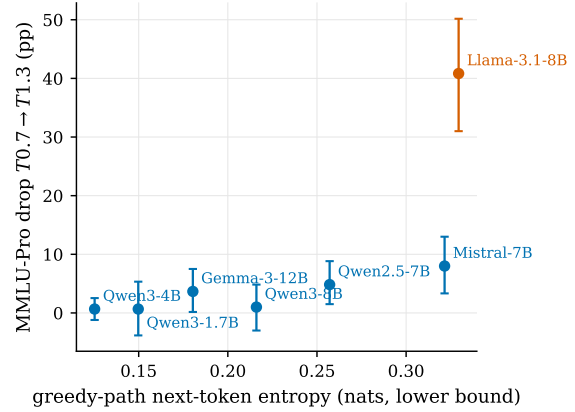


Figure 7: Per-token uncertainty on the model's own greedy path versus the temperature drop (MMLU-Pro, 95% CIs). The rank correlation is  $\rho = +0.96$ ; the Llama–Mistral gap at equal entropy shows flatness alone does not set the magnitude.

## J Mechanism probes

Figure 7 plots greedy-path entropy against the temperature drop. Table 13 consolidates the per-model mechanism evidence of Section 4.6: greedy-path entropy lower bound and flat-position fraction (top-1 < 0.5) from the summary probe; the temperature-scaled beyond-top-20 escape hazard per position at  $T=1.3$  from the raw top-20 arrays; and the perturbation deltas (post-injection minus control over 32 positions, escape hazard and entropy) from the forced-token experiment. Escape columns are  $\times 10^{-3}$ . Every static column ties Llama-3.1 with Mistral; only the perturbation response separates them.

## K Reproducibility

- Inference stack.** llama.cpp at commit 5aba5364 (server build 9456), native /completion endpoint, CUDA builds of the same commit per GPU architecture (sm\_75 for RTX 2080 Ti, sm\_120 for RTX 5070 Ti). Server-side BOS insertion is disabled; the client-rendered chat template owns the special tokens, and each record stores the SHA-256 of the exact rendered prompt.
- Quantizations.** Every quantized checkpoint is a Bartowski imatrix GGUF (one provider, one recipe, so the quantization axis is not confounded by recipe). SHA-256 hashes of all 31 model files are pinned in the repository (DOWNLOADS.md).

Table 10: Accuracy (%) with temperature applied before (left) versus after (right) truncation, pooled over quantization.

| Model           | Method        | temperature-first |      |      | temperature-last |      |      |
|-----------------|---------------|-------------------|------|------|------------------|------|------|
|                 |               | 0.7               | 1.0  | 1.3  | 0.7              | 1.0  | 1.3  |
| <i>GSM8K</i>    |               |                   |      |      |                  |      |      |
| Llama-3.1-8B    | top- <i>p</i> | 81.5              | 78.0 | 63.8 | 80.8             | 78.7 | 73.0 |
| Llama-3.1-8B    | min- <i>p</i> | 81.2              | 78.0 | 72.0 | 78.7             | 81.3 | 76.2 |
| Mistral-7B      | top- <i>p</i> | 28.2              | 31.0 | 34.2 | 29.5             | 31.2 | 29.5 |
| Mistral-7B      | min- <i>p</i> | 32.3              | 28.3 | 28.8 | 29.5             | 32.0 | 30.5 |
| Qwen2.5-7B      | top- <i>p</i> | 92.2              | 91.2 | 90.3 | 90.8             | 91.3 | 90.3 |
| Qwen2.5-7B      | min- <i>p</i> | 90.8              | 90.5 | 91.0 | 92.0             | 90.7 | 91.3 |
| Gemma-3-12B     | top- <i>p</i> | 91.5              | 90.3 | 90.7 | 90.2             | 91.8 | 91.5 |
| Gemma-3-12B     | min- <i>p</i> | 91.5              | 91.3 | 92.3 | 91.2             | 90.8 | 90.3 |
| Qwen3-1.7B      | top- <i>p</i> | 73.8              | 75.0 | 72.7 | 73.0             | 72.0 | 71.3 |
| Qwen3-1.7B      | min- <i>p</i> | 72.2              | 71.7 | 74.0 | 74.7             | 73.3 | 74.5 |
| Qwen3-4B        | top- <i>p</i> | 95.5              | 95.5 | 93.5 | 94.8             | 95.2 | 94.3 |
| Qwen3-4B        | min- <i>p</i> | 95.1              | 94.7 | 92.9 | 94.5             | 93.9 | 93.9 |
| Qwen3-8B        | top- <i>p</i> | 94.0              | 94.3 | 93.7 | 93.8             | 94.0 | 94.0 |
| Qwen3-8B        | min- <i>p</i> | 94.8              | 95.3 | 94.7 | 96.2             | 94.7 | 95.3 |
| <i>MMLU-Pro</i> |               |                   |      |      |                  |      |      |
| Llama-3.1-8B    | top- <i>p</i> | 51.3              | 46.0 | 16.0 | 50.3             | 46.7 | 37.5 |
| Llama-3.1-8B    | min- <i>p</i> | 51.2              | 50.8 | 39.5 | 51.2             | 50.2 | 43.5 |
| Mistral-7B      | top- <i>p</i> | 30.2              | 29.5 | 25.3 | 29.2             | 30.2 | 32.5 |
| Mistral-7B      | min- <i>p</i> | 33.8              | 30.7 | 31.0 | 33.2             | 33.3 | 28.7 |
| Qwen2.5-7B      | top- <i>p</i> | 65.5              | 62.8 | 62.7 | 63.7             | 62.7 | 61.3 |
| Qwen2.5-7B      | min- <i>p</i> | 61.3              | 63.5 | 59.7 | 64.2             | 63.0 | 62.3 |
| Gemma-3-12B     | top- <i>p</i> | 65.7              | 66.7 | 63.8 | 65.8             | 65.0 | 65.2 |
| Gemma-3-12B     | min- <i>p</i> | 65.3              | 67.0 | 65.5 | 65.3             | 63.7 | 63.8 |
| Qwen3-1.7B      | top- <i>p</i> | 47.2              | 47.0 | 47.2 | 47.7             | 46.8 | 43.5 |
| Qwen3-1.7B      | min- <i>p</i> | 47.0              | 47.7 | 46.2 | 45.7             | 46.2 | 49.0 |
| Qwen3-4B        | top- <i>p</i> | 62.8              | 61.3 | 62.8 | 62.9             | 62.7 | 61.7 |
| Qwen3-4B        | min- <i>p</i> | 60.9              | 62.7 | 62.9 | 61.3             | 63.6 | 61.5 |
| Qwen3-8B        | top- <i>p</i> | 60.5              | 61.3 | 60.5 | 60.7             | 59.5 | 60.5 |
| Qwen3-8B        | min- <i>p</i> | 61.0              | 59.2 | 59.7 | 61.3             | 60.7 | 59.3 |

Table 11: Plain-temperature accuracy drop  $T_{0.7} \rightarrow 1.3$  (pp) per quantization level.

| Model           | Q8_0  | Q6_K  | Q4_K_M | Q3_K_M |
|-----------------|-------|-------|--------|--------|
| <i>GSM8K</i>    |       |       |        |        |
| Llama-3.1-8B    | +30.0 | +32.0 | +36.0  | +38.0  |
| Mistral-7B      | -0.7  | -0.7  | +3.3   | +12.0  |
| Qwen2.5-7B      | +4.0  | +2.7  | +2.7   | +6.7   |
| Gemma-3-12B     | +3.3  | +2.7  | +0.0   | +2.0   |
| Qwen3-1.7B      | +5.3  | -6.0  | -0.7   | +0.0   |
| Qwen3-4B        | +2.7  | +0.7  | +6.0   | +2.7   |
| Qwen3-8B        | -2.0  | +1.3  | +2.0   | +0.7   |
| <i>MMLU-Pro</i> |       |       |        |        |
| Llama-3.1-8B    | +38.0 | +46.7 | +38.0  | +40.7  |
| Mistral-7B      | +9.3  | +5.3  | +10.7  | +6.7   |
| Qwen2.5-7B      | +7.3  | +6.0  | +4.0   | +2.0   |
| Gemma-3-12B     | +3.3  | +5.3  | +2.7   | +3.3   |
| Qwen3-1.7B      | +0.7  | +6.0  | -8.0   | +4.0   |
| Qwen3-4B        | -0.7  | +0.0  | +2.0   | +4.0   |
| Qwen3-8B        | +2.7  | -0.7  | +0.0   | +2.0   |

Table 12: Mean distinct parsed answers per item ( $K=3$ , pooled over quantization).

| Model        | GSM8K   |         | MMLU-Pro |         |
|--------------|---------|---------|----------|---------|
|              | $T=0.7$ | $T=1.3$ | $T=0.7$  | $T=1.3$ |
| Llama-3.1-8B | 1.51    | 2.33    | 1.71     | 1.28    |
| Mistral-7B   | 2.21    | 2.48    | 1.97     | 2.21    |
| Qwen2.5-7B   | 1.10    | 1.27    | 1.44     | 1.57    |
| Gemma-3-12B  | 1.12    | 1.16    | 1.33     | 1.45    |
| Qwen3-1.7B   | 1.41    | 1.43    | 1.67     | 1.73    |
| Qwen3-4B     | 1.05    | 1.14    | 1.33     | 1.38    |
| Qwen3-8B     | 1.09    | 1.13    | 1.41     | 1.40    |

- **Decoding.** Exactly one truncation parameter per request; the sampler chain order is pinned per arm and recorded in every record. Sampler values: top-*p* 0.95, top-*k* 40, min-*p* 0.05, top- $n\sigma$  1.0 (each method’s recommended value).

- **Seeding.** Every generation’s seed is derived deterministically from (global seed 0, model, quantization, sampler, temperature, task, item, repetition). Greedy decoding is the deterministic  $T=0$  anchor with  $K=1$ .
- **Tasks.** GSM8K (openai/gsm8k, test split, first 50 items) and MMLU-Pro (TIGER-Lab/MMLU-Pro, test split, first 50 items; stratified replication in Appendix C). Parsers mirror lm-eval-harness; parse failures grade as incorrect and are reported.

Table 13: Mechanism probes per model (Q8, 100 frozen prompts). Escape hazards are at  $T=1.3$ ; drop is the MMLU-Pro plain-temperature  $T_{0.7} \rightarrow 1.3$  accuracy drop (pp), pooled over quantization.

| Model        | $H_{lb}$ | flat% | esc  | $\Delta_{esc}$ | $\Delta H$ | drop  |
|--------------|----------|-------|------|----------------|------------|-------|
| Llama-3.1-8B | 0.329    | 7.1   | 3.26 | +4.20          | +0.110     | +40.8 |
| Mistral-7B   | 0.322    | 6.8   | 3.40 | +1.19          | +0.044     | +8.0  |
| Qwen2.5-7B   | 0.257    | 4.8   | 1.97 | +1.00          | +0.049     | +4.8  |
| Gemma-3-12B  | 0.180    | 3.0   | 1.44 | +1.42          | +0.055     | +3.7  |
| Qwen3-1.7B   | 0.150    | 2.1   | 0.36 | -0.00          | +0.012     | +0.7  |
| Qwen3-4B     | 0.125    | 1.5   | 0.22 | +0.21          | +0.021     | +0.7  |
| Qwen3-8B     | 0.216    | 3.8   | 0.71 | +0.22          | +0.024     | +1.0  |

- **Records.** Each generation is one JSONL record carrying the full identity tuple, the request parameters, the raw output, the parse path, and throughput covariates; runs are resumable by record id. Batched serving ( $\leq 8$  slots) is seed-logged but not bit-exact.
- **Code and data.** The full pipeline, configs, analysis scripts, and every generation record are in an anonymized repository at <https://anonymous.4open.science/r/decoding-robustness-027C>.